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Products

hp 53310A Modulation Domain Analyzer

UNDER CONSTRUCTION

The hp 53310A is essentially a high end universal counter with a graphical display. Variants include the 031 option which adds an internal mixer and local oscillator (LO) (as opposed to a simple pre-scaler) for relatively high resolution frequency measurements up to 2.5 GHz.

The following is a list of the 53310A manual set with some pdfs:

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HP 53310A Quick Start Guide
HP 53310A Operating Reference Manual 53310-90037
HP 53310A Option 031 User's Guide 53310-90025
HP 53310A Programming Quick Reference Guide 53310-34 (online scan credit unknown)
HP 53310A Programming Reference Manual 53310-90038   (online scan credit unknown)
HP 53310A Service Manual 53310-90029
Service Note 53310A-02A
Agilent 53310A Data Sheet 5966-4092E
Agilent 53310A Product Brochure 53310-4093E


The 53310A was sold under the Agilent brand as late as 2001. Pages from the
        Agilent
      2000 catalog include a price list.
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Vintage hp and Agilent 53310A Application Notes give the impression of a relatively low resolution instrument for 0.1% type applications such as plotting the dynamic response of a phased lock loop (PLL) or plotting motor speeds. However, the hp 53310A can also provide high resolution single shot and averaged measurements.

For example, for a 10 MHz signal, a 500 ms sample interval yields 100 (micro) uHz resolution (see graphs 1 and 2 of the data sheet). Averaging successive readings can produce 14 digits (reference oscillator stability over various time frames needs to be considered at such high resolution). By running a 53310A using an external GPS disciplined oscillator (e.g. hp Z3801A) as the external reference clock, 10 MHz signal sources, such as the 10811 series oscillators can be easily tracked over time to a part in 10 to the 10 (10^-10) or better (assuming sufficient GPS disciplined oscillator stability).

Also, as noted in a hp Bench Briefs publication, the instrument is ideal for "setting" precision oscillators using the histogram mode since there is no perceptable delay between successive frequency measurements. The instrument offers great range of sampling rates, number of averaged samples, and display ranges not apparent at first glance. For example, once roughed in, a 100 milli Hz histogram scale (+/- 50 mHz) centered about 10 MHz, proves convenient to adjust a 10811 oscillator to a GPS external frequency reference (connected to the rear panel "EXT REF" BNC input). A 20 ms to 100 ms sample time (interval at center) with 10 to 20 measurements per histogram is one example of a combinations that give relatively rapid updating while tuning an oscillator mechanical adjustment.

Comparing two Z3801A GPS references to each other can be done by connecting one to the EXT REF input and the other to the A or B channel input. A convenient histogram scale is 2 mHz (+/- 1 mHz) centered about 10 MHz. A 0.5 s sample time with 20 sample averaging per acquisition cycle is a good starting point. Two well behaved Z3801As will typically run within 100 uHz ($10^{\wedge -11}$) or better of each other.

Using the histogram mode, the "mean" numeric display shows an average of successive measurements, while the histogram shows the distribution of those measurements. Hist Accumulate continues to fill in a histogram with successive sets of data, roughly analogous to a "persistence" mode.

In the histogram and fast histogram mode, the digital tracking marker displays e.g. blue key "mean" and "standard deviation" provide convenient graphical line marker and digital readouts. With additional averaging by accumulation in the histogram mode, there can be 14 digit frequency displays (the usefulness of such high resolution is of course related to the stability of the external frequency reference over various time periods of interest). While it is very difficult to read so many place on a small graphic number display, a marker can be placed at a center frequency of interest and a second marker set to autotrack a mean value. In this configuration, the instrument displays a delta value, such as -15.6 (milli) mHz. For the delta measurement from a fixed center frequency using tracking marker lines only enable "mean" and not "standard deviation", since with both "mean" and "standard deviation" enabled, the two autotracking markers line up at at +/- one sigma.

Time interval measurements can be made at rates of 1.25 MHz or higher ranging from 20 ns to 1 second. Resolution is limited to about 125 ps without averaging.

Manuals

On fast rise time measurements: This instrument is so under-rated, and under described, I wonder why hp did not do a better job marketing it? It is able to measure sub 1 ns pulse rise times using the +/-TI function. Connect your input to the "A" input and in "Function and Input", select "common". With 50 ohm coupling, use a 20% voltage setting for the A trigger level and an 80% trigger level for the B input (be careful not to exceed the max 5V input, I often use an external 50 ohm terminator instead of the internal terminator just to be safe). Using the time function "vs time" on the histogram menu, look at the time interval (TI vs. time) to be sure a good measurement is being made (i.e. that most of the measurements are in range). For example, for a 10 MHz 1 V pulse, use 200 mV and 800 mV as the trigger levels, with the "common input" options selected (signal coupled to input A). To look at the fastest edges available from our old hp 8082A pulse generator, I ended up using 680 ps center and a 10 ns range (so I could adjust the 8082 "leading edge" knob and watch the effect on the 53310A screen) for the vertical settings and 500 us / div for the time base. I used auto trigger, "edge" sampling. With all data looking good on the time graph, I switched over to the histogram mode. With the deep memory (option 001) you can also go over to the fast histogram mode and select how many measurements to take per histogram update. Turn on "mean" and "std" to see a digital display of the rise time. Histogram accumulate is available if you want to continously average successive measurements. 882 ps rise time is about the best I can do with the old hp 8082A, not bad for a >20 year old pulse generator! With 100,000 measurements, and histogram accumulate on (10M measurements in about 15 seconds at 10 MHz!), I get a consistant 833 ps PDF short term stable to .1 ps! Using the same settings with a 10 MHz (repetition rate), 5 ns wide pulse from an hp 5359A Time synthezer, I get a 20% to 80% rise time of about 1.049 ns, about as expected PDF. I doubt a hp 5372A in the "rise time" mode could do much better than this.

Tech Notes

What goes around, comes around: Here is a bit of irony: (2011) Spectracom and Pendulum have come out with a new software package that allows owners of the Pendulum CNT-91 timer/counter to emulate Agilent Technologies' discontinued Model 53310A MDA. "Pendulum TimeView 3 software from Spectracom turns the Pendulum CNT-91 timer/counter into a real-time measurement analyzer." Test and Measurement World. *See also:* Spectracom's emulation of Agilent 53310A MDA (TimeView 3). Or, you could just buy a used hp or Agilent 53310A for between $300 and $1,500. (Granted both the CNT-91 and software are far more powerful with added functionality.)

For paper records the hp 53310A has a "print" key that sends a print (not a plot) to a GPIB printer. The PrintCapture software package $97 (no affiliation) provides a robust and lightweight ("lightweight" meaning efficiently coded software and not a burden to any system) printer emulator that works great with the hp 53310A (e.g. using a National Instruments GPIB card and a PC). PrintCapture can produce paper copies as well as graphics images, such as for website display. Place the 53310A in "talk only" mode when using the 53310A print key feature with PrintCapture.

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Another capture program, Agilent Screen Capture 2.0, appears to be "freeware" from Agilent. Agilent Screen Capture is another relatively lightweight application that works really well. As compared to PrintCapture, Agilent Screen Capture communicates two-way via GPIB, so instead of "talk only", for this program, select "addressed" in the Utility, HPIB/Print menu. Although more limited in utility and general use than PrintCapture, as to the 53310A, the Agilent Screen Capture 2.0 also makes a super nice print.

Compared to its older cousins, the hp 5371A - 5373A, the user interface is far easier to use and very intuitive once the various available features and settings are found. It is evident that a later generation of programmers and / or newer micros and software tools allowed for a complelely different approach to the user interface. Also, the larger display is far easier to work with. Minor drawbacks are that the 53310A A and B channels cannot simultaneously measure frequency (of course the 53310A can do TI and +/-TI A to B) and there is no built in Allan Variance computation and display as in the 5372A (however, for precision measurements external mixers are still needed, such as for the well known dual mixer techniques described in 537* related hp publications such as Simplify Frequency Stability Measurements with Built-in Allan Variance Analysis (AN 358-12)). With existing 53310A prices, certainly two boxes can be used with external triggering where two simultaneous frequency measurements are needed (not a common application) and other types of analysis are far easier to do today on a PC, Mac, or workstation, after data transfer over GPIB.

The 537*A series did have one very nice feature, a large display of frequency with digits separated 3 at a time for easy determination of decimal places. However as noted above, the 53310 marker feature and digital display suffices to easily read difference values from a reference number. For example, the 5372A might display 10. 000 000 005 4 MHz, compared to the 53310A 10.00000000543 MHz. However with one marker set to 10 MHz and the other to auto-track "mean" the 53310 displays delta 5.43 mHz. We note with pleasure that Agilent brought back the spaced groups of numbers in the new 532* series of universal counters, such as the 53230A 350 MHz Universal Frequency Counter/Timer, 12 digits/s, 20 ps, nice!

Another drawback of the the 537*A series, each having a giant linear power supply transformer, is that they weigh over 65 pounds (with a suggested two person lift, genuine boat anchors!) and take up a large foot print compared to the relatively small and light 53310A. On the other hand, particularly the 5372A was a pretty amazing machine and we recently bought one to have another look at it. Just as with the 53310A, prices vary widely from about $100 to $2000, depending on the seller, condition and options. (*Afternote*: Sadly the hp 5372A arrived with a bent cast aluminum frame and a boot up screen full of hardware errors, way beyond the normal 160 error, and was immediately returned. Because of the 65 pound weight, I would estimate there is only about a 30% to 50% chance of having one of these delivered without significant damage.) (*Afternote*: Trying once more 12/2011, we'll see if I have any better luck this time. nope - seller claims is was dropped during a recent move.) (*Afternote*: 6/2012 okay, maybe one last try for a 5373A, here's to hoping it might arrive okay.) The 5373A (a 5372A with some additional pulse functions. The 5373A cannot take the 040 jitter spectrum analysis option, otherwise it's a 5372A with extra functions) arrived okay (FedEx delivered it box end on the ground, no idea if it was the front or rear panel down, fortunately the seller had installed hard foam and buble wrap. Small miracles do still happen!). After the standard Tadiran battery change (A7 computer board, which allows one to cure the 160 calibration error) and sensitivity calibration (Test 25 with adjustment), it comes on fine. On first use, some years after I looked at the first 5372A, the menus are a little less complicated than I remembered. It is still an instrument that requires more manual study and practice to use effectively, however, in many regards (e.g. direct reading Rt Allan Variance) it certainly is a more sophisticated instrument than the 53310A.

The hp 53310A, 53305A companion software does not at first evaluation appear to give easy control over settings such as sample time and number of samples for each histogram plot. Also, it does not appear to plot the instrument screen, only graphics generated by the hp software. While probably useful in many applications, so far, it appears less friendly for high resolution frequency reference mesurements.

hp 53310As are widely available on the surplus market at erratic and inconsistant sale prices ranging from under $150 to $2,500. Agilent branded units are somewhat less common. List price in 2000 was $11,500 ($19,200 with options). Commercial used test equipment prices also vary widely from $2,000 to $6,000. The 031 option models are said to be popular for some military contractor applications.

The standard instrument has an A input channel (10 Hz to 200 MHz) and a B input channel (10 Hz to 100 MHz).

OPT 001 adds extended memory (not field upgradeable, it's a different mother board ... ***Afternote:*** *I have been told this can be done with some chips and jumpers! Check back, instructions to follow!*)

OPT 010 showing as an aluminum rectangular "bump" on the rear panel adds the standard 10811 oven oscillator (nice to have, but not needed with a GPS, rubidium, or cesium external standard). Unlike other hp counters and generators that can accept a hp 10811 oscillator as a simple plug-in upgrade (in some cases with jumper changes), the hp 53310A requires an entirely different aluminum back panel assembly to support the 010 option. All versions accept an external reference frequency input ("EXT REF IN").

OPT 030 adds a counter style prescaler for a 50 MHz to 2.5 GHz "C" channel. Option 030 units will have a fourth "C" channel BNC at the right most position on the lower right hand side of the front panel (but not the OPT 031 hardline SMA jumper on the lower left rear panel). Here is an example of the front and rear panels of a 53310A 010 030 courtesy of Yixun Test Equipment, Hong Kong, China. Another example, front (This CRT has a somewhat blacker background color than other lighter green colored background colors) and rear

OPT 031 adds an internal mixer and LO for high resolution "C" channel measurements to 2.5 GHz. There is also a rear panel access connector to supply an alternate bench LO source (150

MHz to 2.5 GHz). The digital measurement features are vintage FSK type readings not to be confused with modern in-phase and quatrature (IQ) type measurements (e.g. QAM). OPT 31 is easily identified by the hard line SMA jumper on the lower left hand corner of the back aluminum panel between LO out and LO in. Some units without the option have empty holes there, others a different back panel without the 031 holes or label references. All option 031 units will have a fourth "C" channel BNC at the right most position on the lower right hand side of the front panel. front and rear (option dots not updated in this upgraded unit, also, the keyboard was not updated, missing the FSK blue labels above keys on the right side, functions present, just not labeled as blue shift keys).

Occassionally a used unit shows up for sale with no display (dark CRT). hp contracted the video board to Toshiba or Hitachi and the one fix which is relatively easy to do is to change one capacitor (a non-polarized capacitor, typically with the bottom partly blown out) and a single fuse (both from Mouser Electronics) to put it back in working order in short order. (For those less experienced, danger Hi Voltage!, use best electrical safety practices.) The capacitor is Mouser 140-BPHR50V3.3-RC 50V 3.3uF 6.5Ap-p, the fuse is Mouser 576-0233002.MXP 125V 2A (total repair under $2 not including shipping!). Use care not to over-tighten the small nylon wire tie strap. Just make it snug, not to a point where it deforms the capacitor can. Over-tightening the strap might be what did in some of the original caps. This picture shows the offending capacitor (replaced) and fuse (the unit is partially dissassembled, power supply module is out). Some such units sell for under $150 (although maybe not too many more after this report). Other repairs might be difficult to impossible with complex coated SMT boards. Interpolator errors might simply call for a self calibration routine, or might be an indication of some fatally failed circuit. Spare boards also show up occasionally on the surplus market, however, on occasion, service techs put the "bad" boards back in the replacement board box, so caveat emptor applies. Another relatively easy fix (although very time consuming) is to change out a broken case using a another 53310A case or one from a scrapped hp 54503A oscilloscope (other scopes in that series do not have the right case features). Also, an A2 input trigger light error (FAIL) might simply mean that you need to do an offset (zero) calibration of your input board.

Links

Field Upgrades:

Refer to the Service Manual for more details

OPT 030: It turns out that adding OPT 030 can be done by changing the input A2 board (with the input BNCs). Ideally, you also need a four channel aluminum plate to go over the four holes in the case for the four input BNCs. The unit seems to auto ID the presence of the OPT 030 four channel input board.

OPT031: This one is doable, but a lot of work. In addition to the OPT 031 oscillator board and a special 031 four channel input board (very different from the 030 input board), you need a number of cables. The heavy ribbon power supply cable includes two wires to a power plug for the new oscillator board. Two of the side IDC ribbon cables are completely different with an additional IDC connector for two of the IDC sockets on the new oscillator board. There are also two new coax cables, one mcx connector under the new four channel input board, and one SMB above. There are also new coax cables in the 10 MHz reference lines which go in and out of the new oscillator board. Also, there is are the characteristic gold SMA sockes with the U shaped SMA jumper on the rear panel. Finally, you need a back with the two empty holes for the SMA jacks, meaning you have a "031" ready unit. Also, if the EPROMs are too old (below certain serial numbers), you will need a new set of four EPROMs for the computer board. So, beyond the miracle of a new old stock upgrade kit, why would this ever come up? Because with many damaged units out there (e.g. broken cases), you might find a "031" parts unit to combine with another box to make a nice 031 box with other options.

OPT 001: Extended memory OPT 001 (seems to work ok!)- four 25 ns memory chips U76-U79 JPG from a parted out unit. One user reports that later units had 15 ns memory (believed to be > year 2000, JPG courtesy of Paul Meyer). Otherwise it might be possible to add these chips and re-configure the jumpers to add OPT 001 to a non OPT 001 unit. One person reported field upgrades to add the extended memory were done at his company. Unfortunately, he never got back to us with more details. The manual calls for a different mother board. We *think* the odds are pretty good that a 25 ns can be so upgraded by plugging these chips into the empty sockets and reconfiguring two or three jumpers, however, we do not know for sure. It would seem prudent(?) not to mix memory speeds, e.g. not to use 25 ns RAM for the extended memory where the soldered in base memory is the newer (post 2000?) 15 ns RAM.

OPT 010: Unlike many classic hp instruments were a 10811 type oven controlled oscillator simply plugs into an available socket, the 53310A 010 option requires a very unique back with the OCXO power supply and a box shaped extension which holds the oscillator. For this upgrade, short of the miracle of a new old stock hp upgrade kit, you really need a parts unit that has the 010 option for the special back and OCXO power supply.

Ordering

Summary of Options:

Soft Carrying Case (HP Part Number 1540-1066)
Transit Case (HP Part NumberJ9211-1645)
Service Accessories Kit (HP Part Number 53310-67001)
Rack Mount Kit (HP Part Number 5061-6175)
Rack Mount Slide Kit (HP Part Number 1494-0015)

Option 001 Extended Measurement Memory (4X)
Option 010 High Stability Oven Time Base
Option 030 2.5 GHz Channel C
Option 031 Digital RF Communications
Analysis, High Resolution 2.5 GHz Input
Option 910 Additional set ofUser Documentation
(includes Quick Start Guide and Quick Start Signal
Source)
Option 915 Additional Service Manual
Option 916 Additional set of User Documentation
(excludes Quick Start Guide and Quick Start Signal
Source)
Option W30 Three-year customer retum repair coverage
Option W32 Three-year customer return calibration coverage

hp 53310A STORE

On ocassion, we have hp 53310A parts, boards, and modules for sale:

add actual shipping charges to prices below

power supply board for OPT 010 OCXO units $49

video board $79

all other parts and boards were sold or ended up in a remanufactured unit (see field upgrades above).

Comments welcomed - click here for email.